Calculation Supplement — Document 2

Omnibus Tests

Nine-group Welch ANOVA and restricted Kruskal–Wallis

343 runs / 9 methods / 4 scenarios

Step-by-step statistical analysis
of social navigation in an agricultural digital twin

Purpose. To support the omnibus-analysis calculations. Welch ANOVA from summary statistics is used across nine methods; Kruskal–Wallis is applied only to B2/B3/B4, for which run-level data are available.

Contents

1 Methodological framework

Run-level scalars are available for B2/B3/B4, whereas only aggregates are available for M0–M4/B1. Therefore, nine-group ranks are not reconstructed. The primary omnibus test is heteroscedastic Welch ANOVA:

$$F_W = \frac{\sum_j w_j(\bar{x}_j - \bar{x}_w)^2/(k-1)}{1 + \frac{2(k-2)}{k^2-1}\sum_j \frac{(1-w_j/W)^2}{n_j-1}}, \quad w_j = \frac{n_j}{s_j^2}. \tag{1}$$

Denominator degrees of freedom are approximated by

$$\nu_W = \frac{k^2-1}{3\sum_j(1-w_j/W)^2/(n_j-1)}, \qquad W = \sum_j w_j. \tag{2}$$

The null hypothesis is $H_0 : \mu_1 = \cdots = \mu_9$; rejection means that at least one mean differs but does not identify the responsible pair.

1.1 Kruskal–Wallis and tie correction

For B2/B3/B4, Kruskal–Wallis uses

$$H = \frac{12}{N(N+1)}\sum_{j=1}^{k}\frac{R_j^2}{n_j} - 3(N+1), \qquad df = 2. \tag{3}$$

For ties, $H_c = H/C$, where

$$C = 1 - \frac{\sum_g(t_g^3 - t_g)}{N^3 - N}, \tag{4}$$

and t_g is the size of tied-value group g.

2 Calculation procedure

1. Assemble $(n_j, \bar{x}_j, s_j)$ for every method and metric.
2. Compute $w_j = n_j/s_j^2$, $\bar{x}_w = \sum_j w_j\bar{x}_j/W$, and F_W.
3. Obtain ν_W and the p value from F_{8,ν_W}.
4. For B2/B3/B4, pool observations, assign average ranks, sum R_j, and correct ties.
5. Compare H_c against χ_2^2 and record it without converting the omnibus result into a pairwise claim.

3 Results

4 Interpretation and limitations

Every tested metric differs significantly among methods. This does not identify differing pairs or establish superiority; those questions require the effect sizes in Document 3. The Kruskal–Wallis test is limited to B2/B3/B4 because run-level scalars are available only for those groups.

Table 1: Nine-method Welch ANOVA.

Metric	F_W	df_1	p value
σ_v	421.6	8	1.5×10^{-89}
a_{max}^{num}	79.2	8	$< 10^{-23}$
$T_{personal}$	68.8	8	$< 10^{-23}$
$d^\star$	53.1	8	$< 10^{-23}$
$T_{mission}$	52.5	8	$< 10^{-23}$
N_{stop}	37.7	8	$< 10^{-23}$
L	26.4	8	$< 10^{-23}$

Table 2: Kruskal–Wallis for B2/B3/B4.

Metric	H	df	p value
$d^\star$	84.9	2	$< 10^{-7}$
$T_{personal}$	98.9	2	$< 10^{-7}$
σ_v	86.9	2	$< 10^{-7}$
$T_{mission}$	81.4	2	$< 10^{-7}$
N_{stop}	36.7	2	$< 10^{-7}$

5 Conclusion

Welch is the nine-group analysis, and Kruskal–Wallis is restricted to B2/B3/B4. No individual observations were imputed.

6 Reproducibility control

Implementation must preserve units, sample sizes, and success conditioning; use double precision; record F, H, degrees of freedom, and p before display rounding; and confirm invariance to group ordering. A nine-group rank test requires all run-level scalars.